```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import (
mean_absolute_error,
mean_squared_error,
r2_score
)
# Load Dataset
df = pd.read_csv("Housing.csv")
# Display Dataset
print(df.head())
# Check Missing Values
print(df.isnull().sum())
# Fill Missing Values
df.fillna(df.mean(numeric_only=True), inplace=True)
# Features
X = df[['area', 'bedrooms', 'bathrooms', 'stories', 'parking']]
# Target Variable
y = df['price']
# Train Test Split
X_train, X_test, y_train, y_test = train_test_split(
X,
y,
test_size=0.2,
random_state=42
)
# Create Model
model = LinearRegression()
# Train Model
model.fit(X_train, y_train)
# Prediction
y_pred = model.predict(X_test)
# Evaluation Metrics
mae = mean_absolute_error(y_test, y_pred)
mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)
print("\nModel Evaluation")
print("MAE :", mae)
print("MSE :", mse)
print("R2 Score :", r2)
# Coefficients
print("\nIntercept:")
print(model.intercept_)
print("\nCoefficients:")
for feature, coef in zip(X.columns, model.coef_):
   print(feature, ":", coef)
# Actual vs Predicted
result = pd.DataFrame({
    "Actual": y_test,
    "Predicted": y_pred
})
print(result.head())
# Plot Actual vs Predicted
plt.figure(figsize=(8,5))
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")
plt.show()
```

```
      price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
0  13300000  7420         4          2        3      yes        no       no
1  12250000  8960         4          4        4      yes        no       no
2  12250000  9960         3          2        2      yes        no      yes
3  12215000  7500         4          2        2      yes        no      yes
4  11410000  7420         4          1        2      yes       yes      yes

  hotwaterheating airconditioning  parking prefarea furnishingstatus
0              no             yes        2      yes        furnished
1              no             yes        3       no        furnished
2              no              no        2      yes   semi-furnished
3              no             yes        3      yes        furnished
4              no             yes        2       no        furnished
price               0
area                0
bedrooms            0
bathrooms           0
stories             0
mainroad            0
guestroom           0
basement            0
hotwaterheating     0
airconditioning     0
parking             0
prefarea            0
furnishingstatus    0
dtype: int64

Model Evaluation
MAE : 1127483.352323519
MSE : 2292721545725.3623
R2 Score : 0.5464062355495871

Intercept:
51999.67680883873

Coefficients:
area : 308.86695608764126
bedrooms : 151246.75062952132
bathrooms : 1185731.7137011932
stories : 495100.7626617581
parking : 337660.8302982585
      Actual     Predicted
316  4060000  6.178628e+06
77   6650000  6.370141e+06
360  3710000  3.283148e+06
90   6440000  4.226008e+06
493  2800000  3.409686e+06
```